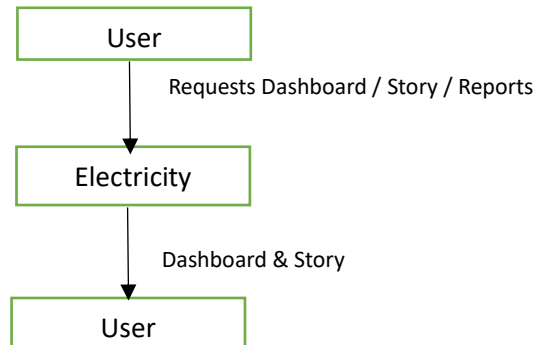


1. Introduction

The Data Flow Diagram (DFD) represents the movement of electricity consumption data within the proposed system. It illustrates how the dataset is collected, stored, processed, visualized, and finally presented to the end user through the Flask web application.

2. Level-0 DFD (Context Diagram)

Create this diagram in **Word** → **Insert** → **Shapes**.

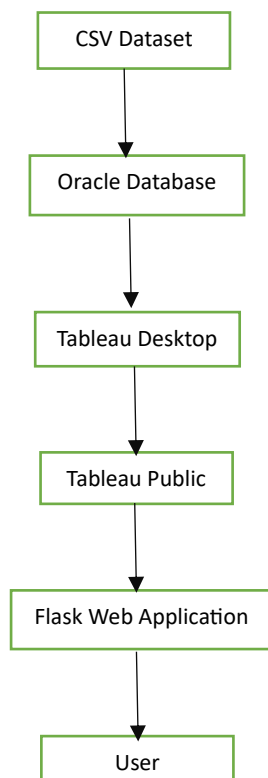


Explanation:

- The **User** interacts with the Electricity Consumption Analysis System.
 - The system processes the request and returns interactive dashboards and stories.
-

3. Level-1 Data Flow Diagram

This is the main DFD.



4. Data Flow Explanation

Step Description

- 1 Electricity consumption dataset is collected in CSV format.
 - 2 Dataset is imported into Oracle Database.
 - 3 SQL queries retrieve the required information.
 - 4 Tableau Desktop creates dashboards and stories.
 - 5 Dashboards are published to Tableau Public.
 - 6 Flask embeds the Tableau Dashboard and Story into the website.
 - 7 Users access interactive reports through the web application.
-

5. Data Stores

Data Store	Description
CSV Dataset	Stores electricity consumption records.
Oracle Database	Stores processed electricity data.
Tableau Public	Stores published dashboards and stories.

6. External Entity

Entity Role

User Views dashboards, applies filters, and analyzes electricity consumption data.

7. Process Description

Process	Description
Data Collection	Collect electricity consumption dataset.
Data Storage	Store data in Oracle Database.
Data Analysis	Execute SQL queries.
Visualization	Create Tableau Dashboard and Story.
Web Integration	Embed Tableau into Flask.

Process	Description
User Access	Users view reports and dashboards.

8. Conclusion

The Data Flow Diagram demonstrates how electricity consumption data moves from the source dataset to the Oracle Database, is processed using SQL, visualized in Tableau, published to Tableau Public, and finally accessed by users through the Flask web application. This flow ensures efficient data processing and interactive visualization.